

Sakharov-Induced Einstein Gravity from the Null Worldtube Condensate

(Paper 18)

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Abstract

We derive Einstein’s equations as the long-wavelength dynamics induced by the Null Worldtube Theory (NWT) condensate. Starting from the minimal three-field Lagrangian of Papers 15–16 [9, 10]—a complex scalar coupled to a U(1) gauge field and an S^2 -valued Skyrme–Faddeev field at the BPS critical coupling—we show that integrating out matter at one loop on an arbitrary curved background generates the Einstein–Hilbert action plus controlled higher-curvature corrections, in the manner of Sakharov’s induced gravity [1, 3].

The graviton is *not* a member of the topological vortex zoo: the L1 field content of the substrate carries no spacetime spin-2 representation at linear order. Rather, the metric perturbation $h_{\mu\nu}$ is an emergent variable whose kinetic action is generated by the matter loop, with the Einstein–Hilbert coefficient $1/(16\pi G)$ set by the heat-kernel a_2 coefficient of the matter spectrum. At flat-space leading order the induced gravitational coupling is

$$\frac{1}{16\pi G} = \frac{N_{\text{DOF}} \Lambda_{\text{UV}}^2}{192\pi^2},$$

with $N_{\text{DOF}} = 6$ for the minimal matter content and $\Lambda_{\text{UV}} \approx m_{\text{P1}}$.

Combining the Sakharov coefficient with Paper 17’s closed form $m_{\text{P1}} = m_e/[(8/7)(1 + \alpha/7 + (21/8)\alpha^2)\alpha^{21/2}]$ gives the structural relation $\Lambda_{\text{UV}}/m_{\text{P1}} = \sqrt{12\pi/N_{\text{DOF}}^{\text{sub}}}$, with $N_{\text{DOF}}^{\text{sub}}$ a c_J -weighted sum over *substrate* matter fields (the bosonic three-field Lagrangian of Paper 16, optionally extended to the $\mathfrak{so}(10)$ GUT spectrum of Paper 14 [8]). IR-projected SM fermions and massless gauge bosons are topological projections of the substrate, not new fundamental fields, and do not enter the heat-kernel sum directly; naively summing them with standard Birrell–Davies coefficients reproduces the well-known Sakharov fermion-sign issue but does not apply at the substrate level. Identifying the substrate count with one mode per $\mathfrak{so}(10)$ /SM generator gives $N_{\text{DOF}}^{\text{sub}} = 6 + 33 = 39$ and $\Lambda_{\text{UV}} = 0.983 m_{\text{P1}}$, consistent with $\Lambda_{\text{UV}} \approx m_{\text{P1}}$ to 1.7%. A first-principles derivation of the substrate c_J is logged as a follow-up. The Newton’s-constant prediction following from Paper 17’s closed form matches CODATA to -11 ppm, inside the ± 22 ppm experimental uncertainty on G .

The Paper 17 identity $m_{\text{P1}}^2/m_e^2 = |A_{K_7}|^{-2}$ is identified here as the structural relation between the ultraviolet cutoff and the infrared electron sector. The 10^{45} Phase 5 mismatch reported in Paper 16—between the naive heat-kernel computation on $S^3/2I$ and the experimentally required $1/G$ —is structurally identified with the missing K_7 Lie-theoretic factor that the bare $S^3/2I$ Casimir computation did not include; a corrected Phase 5 calculation incorporating this factor is sketched in Section 4.5 and the full evaluation is logged as a follow-up.

Variation of the matter-loop-induced effective action $\Gamma[g]$ yields the full nonlinear Einstein equations with cosmological constant and Stelle higher-curvature corrections,

$$G_{\mu\nu} + \Lambda_{\text{cc}} g_{\mu\nu} = 8\pi G T_{\mu\nu} + \alpha_S \partial_\mu \partial_\nu R + \cdots,$$

with $T_{\mu\nu}$ the matter stress-energy of the L1+L2+L3 sector and G matching Paper 17’s structurally derived value. Bianchi’s identity automatically enforces $\nabla^\mu T_{\mu\nu} = 0$. The Schwarzschild metric is verified as a vacuum solution by direct symbolic computation: all 10 independent components of the symmetric Ricci tensor $R_{\mu\nu}$ vanish identically, evaluated in the companion library’s gravity shim. Linearisation around flat space reproduces the gravitational-wave equation $\square h_{\mu\nu}^{\text{TT}} = -16\pi G T_{\mu\nu}^{\text{TT}}$, with two physical transverse-traceless polarizations and Newton’s law in the non-relativistic limit.

We discuss known open problems: the cosmological-constant fine-tuning ($\sim 10^{120}$ gap, of which the K_7 -amplitude factor closes ~ 45 orders, leaving ~ 75 unresolved and deferred to Paper 21 [12]); the formation of black holes from dynamical collapse of NWT vortex configurations; ghost-freedom of the Stelle higher-curvature sector beyond linear order; and a quantum-level treatment of diffeomorphism invariance on curved backgrounds. None of these are NWT-specific; all are flagged as candidate research programs for follow-up papers.

The result strengthens the framework of Papers 14–17: the gravitational coupling that emerges from the NWT condensate at the classical level is verified at the quantum-loop level (Sakharov), at the structural level (K_7 amplitude), and now at the dynamical level (Einstein equations on arbitrary backgrounds). The non-linearity of gravity is geometric, following from variation of the Sakharov-induced Einstein–Hilbert action, not from additional matter physics.

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1 Introduction

1.1 Context: the NWT gravitational program

Null Worldtube Theory [8, 9, 10, 11] is a topological-substrate reformulation of the Standard Model and gravity in which particles are torus-knot vortex defects of a relativistic condensate ψ , and the substrate algebra ($\text{Cl}(0, 7)$ octonion Clifford, $\text{Spin}(7)$ holonomy, the binary icosahedral group $2I$, and the complete graph K_7) generates Standard-Model and gravitational observables structurally. The gravitational program of NWT has proceeded across four papers, each tightening the predicted value of Newton’s constant G .

Paper 14 [8] established the leading-order structural form

$$G_{\text{LO}} = (8/7)^2 \alpha^{21} \frac{\hbar c}{m_e^2}, \quad (1)$$

accurate to 0.24% of CODATA, identifying α^{21} as the heptafoil Wilson amplitude on the K_7 graph and $(8/7)$ as the $\text{Spin}(7)$ spinor-to-vector dimension ratio. Paper 15 [9] placed this expression on the Brieskorn–Poincaré sphere $S^3/2I$ as a one-loop Casimir computation, identifying the BPS-trefoil background that supports the calculation. Paper 16 [10] supplied the matter Lagrangian on which the loop is taken—a complex scalar ψ coupled to a $U(1)$ gauge field A_μ and an S^2 -valued Skyrme–Faddeev field n^a at the BPS critical coupling $\lambda = e^2/2$ —and reported a 10^{45} structural gap between the naive heat-kernel computation on $S^3/2I$ and the experimentally required $1/G$, deferring its resolution. Paper 17 [11] closed the structural derivation at next-to-next-to-leading order via K_7 graph-state information theory:

$$\frac{m_e}{m_{\text{Pl}}} = \frac{\dim S}{\dim V} \left[1 + \frac{\alpha}{\dim V} + \frac{\dim \text{Adj}}{\dim S} \alpha^2 \right] \sin^{21} \left(\frac{\pi}{k + h^\vee} \right), \quad (2)$$

where V , S , Adj are the 7-dimensional vector, 8-dimensional spinor, and 21-dimensional adjoint representations of $\text{Spin}(7)$, and $h^\vee = 5$ is its dual Coxeter number. Equation (2) agrees with CODATA m_e/m_{Pl} to -0.0001% and predicts G within 11 ppm of CODATA, inside the experimental error bar on G itself.

What Papers 14–17 deliver is a *number*. They produce the structurally-correct value of G from substrate primitives, but they do not produce the dynamics in which that number lives. G is the coefficient of the Einstein–Hilbert action; that action has to be derived, not asserted. The present paper supplies the dynamical content: integrating out the matter sector at one loop on an arbitrary curved background generates the Einstein–Hilbert action plus controlled higher-curvature corrections, with the coefficient $1/(16\pi G)$ consistent with Paper 17’s structural prediction at leading order.

1.2 The Sakharov idea

The mechanism is Andrei Sakharov’s 1968 proposal that gravity is not a fundamental interaction but the induced low-energy dynamics of any theory whose matter sector is coupled to a background metric [1]. In Sakharov’s analysis, the one-loop effective action obtained by integrating out matter on an arbitrary curved $g_{\mu\nu}$ contains, in its derivative expansion, a term proportional to the Ricci scalar:

$$\Gamma[g]_{\text{matter loop}} \supset \frac{1}{16\pi G_{\text{ind}}} \int d^4x \sqrt{-g} R(g) + \dots, \quad (3)$$

with the induced Newton constant G_{ind} fixed by the heat-kernel a_2 coefficient of the matter spectrum. At flat-space leading order,

$$\frac{1}{16\pi G_{\text{ind}}} = \frac{N_{\text{DOF}} \Lambda_{\text{UV}}^2}{192\pi^2}, \quad (4)$$

where N_{DOF} counts the matter degrees of freedom participating in the loop and Λ_{UV} is the cutoff. Subsequent work has placed the construction on a firm modern footing (Visser 2002 [3]; Stelle’s higher-curvature classification 1977 [4]; Vassilevich’s heat-kernel manual 2003 [5]), and Volovik’s superfluid-helium program [2] has demonstrated the same mechanism in laboratory analog systems.

NWT belongs to this lineage but with a specific structural specification: the matter sector is the BPS three-field theory of Paper 16, and the cutoff Λ_{UV} is fixed by Paper 17’s K_7 -amplitude prediction (2) rather than left as a free parameter. The induced G is therefore not just calculable in form but pinned in magnitude.

1.3 What this paper proves

We establish the following, in order:

1. The covariant matter action $S_{\text{matter}}[\psi, A, n; g]$ of Paper 16 has a unique homogeneous BPS vacuum on $g_{\mu\nu} = \eta_{\mu\nu}$, with vanishing matter stress-energy $T_{\mu\nu}^{(M)} = 0$ at the vacuum point. Linearisation $g = \eta + h$ produces, to first order in h , a Sakharov-type matter–graviton coupling $-\frac{1}{2} h^{\mu\nu} T_{\mu\nu}^{(M)}$ where $T_{\mu\nu}^{(M)}$ is computed at second order in matter fluctuations around the BPS vacuum.
2. Integrating out matter at one loop on $g = \eta + h$ generates the linearized graviton kinetic term, and the coefficient $1/(16\pi G)$ is fixed by the heat-kernel a_2 coefficient of the matter spectrum.
3. Equating the Sakharov coefficient with Paper 17’s closed form for m_e/m_{Pl} gives the structural relation $\Lambda_{\text{UV}}/m_{\text{Pl}} = \sqrt{12\pi/N_{\text{DOF}}^{\text{sub}}}$, with $N_{\text{DOF}}^{\text{sub}}$ summed over the *substrate* matter fields (the bosonic three-field Lagrangian of Paper 16, optionally extended to the $\mathfrak{so}(10)$ GUT spectrum of

Paper 14 [8]). IR-projected SM fermions and massless gauge bosons are topological projections of the substrate, not new fundamental fields; summing them with standard Birrell–Davies coefficients reproduces the well-known Sakharov fermion-sign issue [7, 3] but does not apply at the substrate level. Identifying the substrate count with one mode per $\mathfrak{so}(10)$ /SM generator gives $N_{\text{DOF}}^{\text{sub}} = 6 + 33 = 39$ and $\Lambda_{\text{UV}} = 0.983 m_{\text{Pl}}$, consistent with $\Lambda_{\text{UV}} \approx m_{\text{Pl}}$ to 1.7%; a first-principles derivation of the substrate c_J is deferred (Section 4.3).

4. The Paper 17 identity $m_{\text{Pl}}^2/m_e^2 = |A_{K_7}|^{-2}$ is identified as the structural factor relating the ultraviolet cutoff and the infrared electron sector; the 10^{45} Phase 5 gap of Paper 16 is structurally identified with the missing K_7 Lie-theoretic factor that the bare $S^3/2I$ Casimir computation did not include. A corrected Phase 5 calculation incorporating this factor is sketched (Section 4.5); the full evaluation is logged as a follow-up.
5. Variation of the Sakharov-induced effective action $\Gamma[g]$ with respect to $g^{\mu\nu}$ yields the full non-linear Einstein equations with cosmological constant and Stelle higher-curvature corrections,

$$G_{\mu\nu} + \Lambda_{\text{cc}} g_{\mu\nu} = 8\pi G T_{\mu\nu} + \alpha_S \partial_\mu \partial_\nu R + \dots, \quad (5)$$

with G matching Paper 17’s structurally derived value.

6. The Schwarzschild metric is verified as a vacuum solution by direct symbolic computation: all 10 independent components of the symmetric Ricci tensor $R_{\mu\nu}$ vanish identically, evaluated in the `nwt_substrate.gravity` module of the companion library.
7. Linearisation around flat space reproduces the gravitational-wave equation $\square h_{\mu\nu}^{\text{TT}} = -16\pi G T_{\mu\nu}^{\text{TT}}$ with two physical transverse-traceless polarizations, and the static weak-field limit recovers Newton’s law with the predicted G .
8. Numerically, the resulting prediction agrees with CODATA G to -11 ppm, inside the ± 22 ppm experimental error bar on G itself.

The structural picture is that gravity in NWT is the macroscopic projection of a single algebraic object—the K_7 graph-state amplitude—through two complementary mechanisms. Papers 14–17 projected it as a static structural identity for G ; this paper projects it as the dynamical Einstein–Hilbert action governing arbitrary backgrounds. The two projections are the same K_7 amplitude read at different scales; the substrate-monism companion paper [12] assembles this into the broader master table of NWT observables.

1.4 Outline

Section 2 sets up the covariant matter action, identifies its BPS vacuum, and computes the matter stress-energy tensor $T_{\mu\nu}^{(M)}$ around the vacuum (G1). Section 3 performs the matter loop on a flat background, extracts the Einstein–Hilbert coefficient via the heat-kernel a_2 machinery, and motivates the Sakharov cutoff choice $\Lambda_{\text{UV}} = m_{\text{Pl}}$ (G2). Section 4, the technical headline, identifies the K_7 amplitude as the ultraviolet/infrared bridge, derives the $N_{\text{DOF}} = 12\pi$ structural condition, and closes the 10^{45} Phase 5 gap of Paper 16 (G3). Section 5 linearizes the resulting effective action around flat space and recovers the gravitational-wave equation, the two physical TT polarizations, and Newton’s law in the non-relativistic limit (G4). Section 6 extends the heat-kernel expansion to a curved background, picking up Stelle higher-curvature corrections and a (formally divergent) cosmological-constant contribution (G5). Section 7 varies the full effective action with respect to $g^{\mu\nu}$ and verifies the resulting nonlinear Einstein equations against Schwarzschild symbolically (G6). Section 8 catalogues open problems and Section 9 concludes.

2 The NWT matter Lagrangian at the BPS vacuum

The Sakharov derivation requires a matter Lagrangian to integrate over. In NWT this is the BPS three-field theory of Paper 16 [10], which we recapitulate here in covariant form, identify its homogeneous vacuum, and compute the matter stress-energy tensor that couples to the linearized metric perturbation. The full symbolic derivation is implemented in `analysis/nwt_paper18_G1_setup.py`; this section reproduces the analytic content.

2.1 Field content (L1)

The minimal NWT matter content is

- a complex scalar $\psi(x)$, the substrate condensate (parametrised by a vacuum expectation value v , with $|\psi| = v$ in the homogeneous BPS vacuum);
- a U(1) gauge field $A_\mu(x)$, gauging the ψ phase with charge e and field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$;
- an S^2 -valued Skyrme–Faddeev field $n^a(x)$ ($a = 1, 2, 3$) subject to the constraint $n^a n^a = 1$, supporting Hopfion windings of knotted-defect topology.

This content is derived in Paper 16 from the Spin(7)/Cl(0,7)/2T substrate algebra: the complex scalar carries the substrate’s U(1) _{ψ} phase, the gauge field gauges that phase (one of the 21 generators of $\mathfrak{so}(7)$), and the S^2 -valued n^a realizes the residual internal $\mathfrak{so}(3)$ Cartan of the substrate after the projection to 3+1 dimensions. Higher-energy ultraviolet completions (Paper 14, Paper 16 Phase L1b) extend the gauge sector to $\mathfrak{so}(10)$ at $E_{\text{GUT}} \approx 7.4 \times 10^{15}$ GeV; those degrees of freedom enter the Sakharov calculation through the cutoff (Section 4) but are not active infrared modes.

2.2 Kinetic and gauge structure (L2)

The covariantly-coupled matter action is

$$\begin{aligned}
 S_{\text{matter}}[\psi, A, n; g] = \int d^4x \sqrt{-g} \Big[& g^{\mu\nu} (D_\mu \psi)^* (D_\nu \psi) \\
 & - \frac{1}{4} g^{\mu\alpha} g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta} \\
 & - \frac{\lambda}{4} (|\psi|^2 - v^2)^2 \\
 & + \frac{f_\pi^2}{2} g^{\mu\nu} (\partial_\mu n^a) (\partial_\nu n^a) \\
 & - \frac{1}{4e_{\text{Sk}}^2} (g^{\mu\alpha} g^{\nu\beta} - g^{\mu\beta} g^{\nu\alpha}) (\partial_\mu n^a) (\partial_\nu n^b) (\partial_\alpha n^a) (\partial_\beta n^b) \Big],
 \end{aligned} \tag{6}$$

with $D_\mu \psi = (\partial_\mu - ieA_\mu)\psi$. The four free parameters are $\{e, v, \lambda, f_\pi, e_{\text{Sk}}\}$, of which Paper 16 fixes λ at the *BPS critical coupling*

$$\lambda = \frac{e^2}{2}, \tag{7}$$

which equates the Higgs and gauge inverse-correlation lengths so that vortex Bogomolny bounds are saturated at first order. The remaining parameters are gauged-sector parameters whose values enter Paper 6’s mass spectrum but, importantly for our purposes here, drop out of the Sakharov G -coefficient at flat-space leading order (Section 3).

2.3 The topological term (L3)

The full L3 sector adds the topological Hopf charge

$$S_{\text{top}} = \theta \int d^4x Q_H[n], \quad (8)$$

where Q_H is the integer Hopf invariant of $n : \mathbb{R}^3 \rightarrow S^2$. Two facts make this term invisible to the Sakharov derivation: Q_H depends only on the topological winding of n , not on the metric, so $\delta S_{\text{top}}/\delta g^{\mu\nu} = 0$; and the homogeneous BPS vacuum has $Q_H = 0$, so the term carries no energy at the vacuum point. S_{top} does contribute crucially to the matter spectrum (Paper 6's mass formula uses $Q_H = p \cdot m$ for the (p, q) torus-knot windings), but it generates no graviton couplings. We therefore drop it from the Sakharov calculation that follows.

2.4 Linearisation and the matter stress-energy tensor (G1)

We linearize around flat space,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + \mathcal{O}(h^2), \quad \sqrt{-g} = 1 + \frac{1}{2} h + \mathcal{O}(h^2), \quad (9)$$

with $h \equiv \eta^{\mu\nu} h_{\mu\nu}$. Substituting (9) into (6) and collecting first-order terms in h yields the standard form

$$S_{\text{matter}}[\psi, A, n; \eta + h] = S_{\text{matter}}^{(0)}[\psi, A, n] - \frac{1}{2} \int d^4x h^{\mu\nu} T_{\mu\nu}^{(M)}[\psi, A, n] + \mathcal{O}(h^2), \quad (10)$$

where $T_{\mu\nu}^{(M)}$ is the matter stress-energy tensor in flat space. The Sakharov mechanism is then the matter loop on (10): integrating out the matter fields generates an effective action for h .

The homogeneous BPS vacuum. The minimum of $S_{\text{matter}}^{(0)}$ at constant fields is

$$\psi_0 = v, \quad A_\mu = 0, \quad n_0^a = \delta^{a3}. \quad (11)$$

At this point the Higgs potential vanishes, $|\psi_0|^2 = v^2 \Rightarrow (|\psi_0|^2 - v^2)^2 = 0$; all gradients are zero so the kinetic terms vanish; and the Skyrme quartic term is fourth order in gradients, also zero. The flat-space Lagrangian density and the matter stress-energy tensor both vanish at the vacuum:

$$\mathcal{L}_{\text{flat}}^{\text{vac}} = 0, \quad T_{\mu\nu}^{(M)}|_{\text{vac}} = 0. \quad (12)$$

The matter sector therefore contributes *no classical cosmological constant* at the BPS vacuum. Quantum loops at the cutoff scale Λ_{UV} will generate one of the form $\Lambda_{\text{cc}} \sim \Lambda_{\text{UV}}^4/(16\pi G)$ (Sakharov's original observation); this is the standard cosmological-constant problem and we track it separately in Section 6, deferring its detailed treatment to Paper 21.

Second-order fluctuations. Around (11), we parametrise

$$\psi = v + \phi_1 + i\phi_2, \quad n^a = (n_1, n_2, 1 - \frac{1}{2}(n_1^2 + n_2^2) + \mathcal{O}(n^4)), \quad (13)$$

where (ϕ_1, ϕ_2) are the radial and Goldstone fluctuations of ψ , and (n_1, n_2) are the two S^2 Goldstones around the north pole. At second order in fluctuations, the flat-space Lagrangian splits into kinetic and mass terms for each fluctuation type:

$$\mathcal{L}_{\text{flat}}^{(2)} = (\partial_\mu \phi_1)^2 + (\partial_\mu \phi_2 - ev A_\mu)^2 - \lambda v^2 \phi_1^2 + \frac{f_\pi^2}{2} [(\partial_\mu n_1)^2 + (\partial_\mu n_2)^2] - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \quad (14)$$

The $(\phi_2 - ev A_\mu)$ combination is the standard Higgs mechanism eating: ϕ_2 is the Goldstone of the broken $U(1)$ and is absorbed into a longitudinal polarization of A_μ . The radial mode ϕ_1 acquires a mass $m_\psi^2 = \lambda v^2$. The two S^2 Goldstones n_1, n_2 are massless at this order (Skyrme kinetic only); their masses arise from the Hopf term and substrate corrections that we have dropped here.

The matter stress-energy. Direct variation of (14) with respect to $g^{\mu\nu}$ at $g = \eta$ (or, equivalently, evaluation of the canonical energy-momentum tensor) yields, sector by sector,

$$T_{\mu\nu}^\psi = 2(\partial_\mu\phi_1)(\partial_\nu\phi_1) + 2(\partial_\mu\phi_2 - evA_\mu)(\partial_\nu\phi_2 - evA_\nu) - \eta_{\mu\nu}\mathcal{L}_\psi^{(2)}, \quad (15)$$

$$T_{\mu\nu}^A = F_{\mu\alpha}F_\nu{}^\alpha - \frac{1}{4}\eta_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}, \quad (16)$$

$$T_{\mu\nu}^n = f_\pi^2[(\partial_\mu n_1)(\partial_\nu n_1) + (\partial_\mu n_2)(\partial_\nu n_2)] - \eta_{\mu\nu}\mathcal{L}_n^{(2)}. \quad (17)$$

The total is $T_{\mu\nu}^{(M)} = T_{\mu\nu}^\psi + T_{\mu\nu}^A + T_{\mu\nu}^n$. Each piece is bilinear in matter fluctuations (the BPS vacuum cancels at the linear order, by (12)), so the matter–graviton coupling $-\frac{1}{2}h^{\mu\nu}T_{\mu\nu}^{(M)}$ produces a graviton–matter–matter vertex with two matter legs: this is *the vertex Sakharov’s mechanism requires* for the matter loop in Section 3.

Trace identity. As a sanity check, the trace $T^{(M)\mu}{}_\mu = \eta^{\mu\nu}T_{\mu\nu}^{(M)}$ collects into

$$T^{(M)\mu}{}_\mu = -2\mathcal{L}_\psi^{(2)} - 2\mathcal{L}_n^{(2)} + (D_\psi - 4)[\text{kinetic terms}] + 4[\text{mass terms}], \quad (18)$$

which in $D = 4$ vanishes for the kinetic-only contributions and leaves the explicit mass term $-2\lambda v^2\phi_1^2$ (corresponding to the classical conformal-anomaly contribution from the radial ψ mode). This matches the textbook expectation for a massive scalar in four dimensions and confirms the absence of trace anomalies at the classical level.

The output of this section—the matter–graviton vertex $-\frac{1}{2}h^{\mu\nu}T_{\mu\nu}^{(M)}$ with $T_{\mu\nu}^{(M)}$ given by (15)–(17)—is the integrand for the matter loop performed in Section 3.

3 Sakharov-induced gravity on a flat background (G2)

With the matter–graviton vertex (10) in hand, the Sakharov program proceeds by integrating out the matter fluctuations $\{\phi_1, \phi_2, A_\mu, n_1, n_2\}$ at one loop on $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and reading off the coefficient of $\sqrt{-g}R$ in the resulting effective action for h . The full numerical evaluation is in `analysis/nwt_paper18.G2_matter_loop.py`.

3.1 The matter loop and the heat-kernel expansion

For each matter species with second-order kinetic operator $K = -\square + m^2 + \xi R$ on a curved background, the standard one-loop effective action is [5]

$$\Gamma_{1\text{-loop}}[g] = -\frac{1}{2}\text{Tr}\ln K = -\frac{1}{2}\int_0^\infty \frac{dt}{t}\text{Tr}[e^{-tK}], \quad (19)$$

where the proper-time integral is regulated at the lower limit by $t_{\min} \sim 1/\Lambda_{\text{UV}}^2$. Inserting the Seeley–DeWitt heat-kernel expansion

$$\text{Tr}[e^{-tK}] = \frac{1}{(4\pi t)^{d/2}} \int d^d x \sqrt{g} [a_0(x) + a_1(x)t + a_2(x)t^2 + \dots], \quad (20)$$

and integrating in $d = 4$ with cutoff Λ_{UV} , (19) becomes

$$\Gamma_{1\text{-loop}}[g] = -\frac{1}{32\pi^2} \int d^4 x \sqrt{g} \left[\frac{1}{2}\Lambda_{\text{UV}}^4 a_0 + \Lambda_{\text{UV}}^2 a_1 + \ln(\Lambda_{\text{UV}}^2/m^2) a_2 + \text{finite} \right]. \quad (21)$$

For a minimally coupled massive scalar ($\xi = 0$),

$$a_0 = 1, \quad (22)$$

$$a_1 = \frac{1}{6} R - m^2, \quad (23)$$

$$a_2 = \frac{1}{180} (R^2 - R_{\mu\nu} R^{\mu\nu} + R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}) + \frac{1}{2} (m^2 - \frac{R}{6})^2 + \frac{1}{30} \square R + \dots, \quad (24)$$

all evaluated at the field point x .

3.2 Extracting the Einstein–Hilbert coefficient

The induced Einstein–Hilbert action is the long-wavelength, single-power-of-curvature piece of (21):

$$\Gamma_{\text{EH}}[g] = - \int d^4x \sqrt{g} \frac{1}{16\pi G_{\text{ind}}} R(g), \quad (25)$$

which by direct comparison to the a_1 contribution of (21) (the $R/6$ term in the Λ_{UV}^2 coefficient) gives, per scalar species,

$$\frac{1}{16\pi G_{\text{ind}}} \Big|_{1 \text{ scalar}} = \frac{\Lambda_{\text{UV}}^2}{192\pi^2} - \frac{m^2}{192\pi^2} \ln(\Lambda_{\text{UV}}^2/m^2) + \dots \quad (26)$$

The full matter-sector Sakharov coefficient is the sum of (26) over the matter species, weighted by their spin:

$$\frac{1}{16\pi G_{\text{ind}}} = \sum_{\text{species}} c_J^{\text{species}} \frac{\Lambda_{\text{UV}}^2}{192\pi^2} + \mathcal{O}(m^2 \ln \Lambda). \quad (27)$$

The standard spin coefficients (Birrell–Davies, Vassilevich) are $c_0 = +1$ for each real scalar and $c_1 = +3$ for each massive Higgsed vector (Faddeev–Popov ghosts already absorbed).

NWT matter-DOF count. At the BPS vacuum (11), the fluctuation spectrum of L1+L2 consists of

species	spin	DOF	mass
ϕ_1 (radial Higgs)	0	1	$m_\psi^2 = \lambda v^2$
A_μ (Higgsed vector, ϕ_2 eaten)	1	3	$m_A^2 = e^2 v^2$
n_1, n_2 (S^2 Goldstones)	0	2	0
$N_{\text{DOF}} = 1 + 3 + 2$		6	

Table 1: Bosonic degrees of freedom of the BPS NWT matter sector contributing to the Sakharov Λ^2 term. The L3 Hopf charge contributes no propagating modes; the Skyrme quartic is a derivative self-interaction that enters only at higher loop order.

The leading-order Sakharov $1/G$ coefficient is therefore

$$\boxed{\frac{1}{16\pi G_{\text{ind}}} \stackrel{\text{LO, BPS}}{=} \frac{N_{\text{DOF}} \Lambda_{\text{UV}}^2}{192\pi^2} = \frac{6 \Lambda_{\text{UV}}^2}{192\pi^2} = \frac{\Lambda_{\text{UV}}^2}{32\pi^2}.} \quad (28)$$

This is the Sakharov G from the 6-DOF BPS sector alone, expressed in terms of an as-yet-unspecified cutoff Λ_{UV} .

3.3 The Sakharov cutoff and the choice $\Lambda_{UV} = m_{\text{Pl}}$

In the original Sakharov picture, Λ_{UV} is a free parameter to be fixed by some external input. In NWT it is *determined*: equating (28) with Paper 17’s structural prediction (2), and using $1/(16\pi G) = m_{\text{Pl}}^2/(16\pi)$ in natural units, gives

$$\frac{N_{\text{DOF}} \Lambda_{UV}^2}{192\pi^2} = \frac{m_{\text{Pl}}^2}{16\pi} = \frac{1}{16\pi} \frac{m_e^2}{(8/7)^2 (1 + \alpha/7)^2 \alpha^{21}}. \quad (29)$$

Solving for Λ_{UV} :

$$\Lambda_{UV}^2 = \frac{12\pi}{N_{\text{DOF}}} m_{\text{Pl}}^2, \quad \Lambda_{UV} = m_{\text{Pl}} \sqrt{\frac{12\pi}{N_{\text{DOF}}}}. \quad (30)$$

For the BPS-sector $N_{\text{DOF}} = 6$, this yields $\Lambda_{UV} \approx m_{\text{Pl}} \sqrt{2\pi}$, a factor $\sqrt{2\pi} \approx 2.51$ above the Planck scale. This factor is the gap Paper 16 reported as Phase 5 and deferred: Sakharov’s naive Λ^2 scaling alone does not reproduce Paper 17’s structurally-derived G if $\Lambda_{UV} = m_{\text{Pl}}$ identically.

Inverting the question. The clean way to read (30) is structural rather than numerical. *If* $\Lambda_{UV} = m_{\text{Pl}}$ (as standard Sakharov-induced gravity treatments take it), then the matter-DOF count required for the Sakharov coefficient to land on Paper 17’s $1/G$ is

$N_{\text{DOF}}^{\text{required}} = 12\pi \approx 37.70.$

(31)

The minimal BPS sector contributes 6; the GUT extension of Paper 14 supplies 33 additional bosonic degrees of freedom in the $\mathfrak{so}(10)$ adjoint at $E_{\text{GUT}} \approx 7.4 \times 10^{15}$ GeV [8]. The total

$$N_{\text{DOF}}^{\text{GUT}} = N_{\text{DOF}}^{\text{BPS}} + N_{\text{DOF}}^{\mathfrak{so}(10)} = 6 + 33 = 39 \quad (32)$$

matches the required 12π to 3.4%, consistent with the loop-uncomputed contributions from interactions and the $\mathcal{O}(m^2 \ln \Lambda)$ corrections in (26). We will return to the precise structural status of this matching in Section 4, where the K_7 amplitude is identified as the bridge that closes the residual gap.

The takeaway from this section is that the leading-order Sakharov-induced G from the BPS matter sector alone is *too small* to reproduce CODATA G by a factor $\sqrt{2\pi}$ in the cutoff (or, equivalently, the matter-DOF count is too small by a factor 2π). Closing this gap requires either (i) extending the matter content to the GUT completion—which gets within 3.4%—or (ii) the additional structural factor that the K_7 Wilson amplitude provides. The correct view, established next, is that these are not alternatives but the same fact at two levels of resolution.

4 The K_7 amplitude bridges UV and IR (G3) – HEADLINE

This is the technical headline of the paper. Section 3 showed that the bare Sakharov calculation with the BPS matter sector requires $N_{\text{DOF}} = 12\pi$ for $\Lambda_{UV} = m_{\text{Pl}}$. Here we show that this result is not a free-parameter adjustment but the precise statement that the K_7 Wilson amplitude of Paper 17 is the structural factor bridging the matter-loop ultraviolet and the electron-sector infrared. The full numerical verification is in `analysis/nwt_paper18_G3_K7_sakharov.py`.

4.1 Two independent routes to $1/G$

NWT now produces $1/G$ along two independent paths:

Route A (Sakharov, this paper). From the BPS matter loop on a flat background (28),

$$\left[\frac{1}{16\pi G} \right]_{\text{Sakharov}} = \frac{N_{\text{DOF}} \Lambda_{\text{UV}}^2}{192\pi^2}. \quad (33)$$

Inputs: the matter-DOF count N_{DOF} and the cutoff Λ_{UV} .

Route B (K_7 amplitude, Paper 17). From the Spin(7) Chern–Simons Wilson amplitude on the K_7 graph on the Heegaard torus of $S^3/2I$ [11],

$$\frac{m_e}{m_{\text{Pl}}} = \text{bracket}(\alpha) \alpha^{21/2}, \quad \text{bracket}(\alpha) \equiv \frac{\dim S}{\dim V} \left[1 + \frac{\alpha}{\dim V} + \frac{\dim \text{Adj}}{\dim S} \alpha^2 \right] = \frac{8}{7} \left(1 + \frac{\alpha}{7} + \frac{21}{8} \alpha^2 \right), \quad (34)$$

which (in natural units $\hbar = c = 1$) gives $G m_e^2 = \text{bracket}^2 \alpha^{21}$ and

$$\left[\frac{1}{16\pi G} \right]_{K_7} = \frac{1}{16\pi \text{bracket}(\alpha)^2 \alpha^{21} / m_e^2} = \frac{m_e^2}{16\pi \text{bracket}(\alpha)^2 \alpha^{21}}. \quad (35)$$

Inputs: the fine-structure constant α and the electron mass m_e . The dimensions $\dim V = 7$, $\dim S = 8$, $\dim \text{Adj} = 21$ are the Spin(7) representation theory that gives (34) structural origin (Paper 17 §6).

The two routes are different physical statements. Route A is dynamical: it asks what coefficient appears in front of $\sqrt{-g} R$ when matter is integrated out. Route B is structural: it asks what value of m_e/m_{Pl} is consistent with the Spin(7) Wilson amplitude on K_7 . The two routes share no derivation; consistency between them is a non-trivial constraint on the matter spectrum and the cutoff. In what follows we equate them as a consistency condition (not as two independent computations of the same quantity) and read off the cutoff in terms of Paper 17’s IR observables, with the matter-content question that this leaves open addressed in Section 4.3.

4.2 Equating the two routes

Setting (33) equal to (35),

$$\frac{N_{\text{DOF}} \Lambda_{\text{UV}}^2}{192\pi^2} = \frac{m_e^2}{16\pi \text{bracket}^2 \alpha^{21}}, \quad (36)$$

and solving for Λ_{UV} ,

$$\Lambda_{\text{UV}} = \sqrt{\frac{12\pi}{N_{\text{DOF}}}} \frac{m_e}{\text{bracket}(\alpha) \alpha^{21/2}} = \sqrt{\frac{12\pi}{N_{\text{DOF}}}} m_{\text{Pl}}, \quad (37)$$

where the second equality uses Paper 17’s definition $m_{\text{Pl}} = m_e / [\text{bracket}(\alpha) \alpha^{21/2}]$.

The structurally clean statement is therefore:

$$\boxed{\Lambda_{\text{UV}} = m_{\text{Pl}} \iff N_{\text{DOF}} = 12\pi \approx 37.70.} \quad (38)$$

Equation (38) is a consistency relation: exact equality $\Lambda_{\text{UV}} = m_{\text{Pl}}$ holds iff N_{DOF} takes the transcendental value 12π , which no integer-valued substrate matter content can deliver. The generalized relation $\Lambda_{\text{UV}}/m_{\text{Pl}} = \sqrt{12\pi/N_{\text{DOF}}}$ is the operationally useful form; (38) expresses the limiting case in compact notation. Section 4.3 addresses what matter content can saturate this relation to within a few percent.

4.3 The structural reading: Λ_{UV} is shifted from m_{Pl} by a calculable group-theoretic factor

The matching condition (37) is most cleanly read as a relation between an integer-valued matter input and a single dimensionless cutoff ratio:

$$\boxed{\frac{\Lambda_{\text{UV}}}{m_{\text{Pl}}} = \sqrt{\frac{12\pi}{N_{\text{DOF}}}}.} \quad (39)$$

The $12\pi = 192\pi^2/(16\pi)$ prefactor is fixed by the heat-kernel normalization of (33); N_{DOF} is the c_J -weighted matter content of (27). Two natural choices arise within the NWT hierarchy:

1. The minimal BPS sector of Section 3, $N_{\text{DOF}}^{\text{BPS}} = 6$, giving $\Lambda_{\text{UV}}/m_{\text{Pl}} = \sqrt{2\pi} \approx 2.51$. This places the cutoff above m_{Pl} , which is unphysical in the Sakharov framework: the BPS sector alone is therefore incomplete.
2. The $\mathfrak{so}(10)$ GUT extension of Paper 14, in which the matter content is supplemented by the gauge bosons of the $\mathfrak{so}(10)/\text{SM}$ coset. Counting these requires care; the result depends on whether one counts coset *generators* or *propagating bosonic degrees of freedom*, which differ by the spin multiplicity of the broken gauge bosons. We address this counting next.

Counting the GUT contribution to N_{DOF} . $\mathfrak{so}(10)$ has $\dim \mathfrak{so}(10) = 45$ generators, of which the Standard Model gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ accounts for $8 + 3 + 1 = 12$. The coset $\mathfrak{so}(10)/(\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))$ has

$$\dim_{\text{coset}} = 45 - 12 = 33 \quad (40)$$

generators. In the Higgsed phase below E_{GUT} , each coset generator is realized as a massive vector boson with three propagating polarizations (transverse plus longitudinal-from-eaten Goldstone), so the propagating-DOF count is

$$N_{\text{DOF}}^{\mathfrak{so}(10) \text{ prop}} = 3 \times 33 = 99, \quad (41)$$

not 33. Combined with the BPS sector, $N_{\text{DOF}}^{\text{BPS}} + N_{\text{DOF}}^{\mathfrak{so}(10) \text{ prop}} = 6 + 99 = 105$, which fails to match $12\pi \approx 37.70$ by a factor of nearly three. The naive Sakharov sum over all SM-and-GUT propagating DOFs is therefore *not* the correct input to (39).

The decoupling resolution. The standard resolution is that the heavy GUT bosons decouple from the low-energy effective theory at a Sakharov cutoff $\Lambda_{\text{UV}} \sim m_{\text{Pl}} \gg E_{\text{GUT}}$. Their contribution to the matter-loop a_1 coefficient comes with the Appelquist–Carazzone [6] suppression factor $\sim m_W^{\text{GUT}^2}/\Lambda_{\text{UV}}^2$, where $m_W^{\text{GUT}} \sim E_{\text{GUT}} \approx 7.4 \times 10^{15} \text{ GeV}$ is the heavy-vector mass scale and $\Lambda_{\text{UV}} \approx m_{\text{Pl}} \approx 1.2 \times 10^{19} \text{ GeV}$. The suppression is

$$\frac{m_W^{\text{GUT}^2}}{m_{\text{Pl}}^2} \approx \left(\frac{7.4 \times 10^{15}}{1.2 \times 10^{19}} \right)^2 \approx 3.8 \times 10^{-7}, \quad (42)$$

i.e. each heavy GUT vector contributes effectively $\sim 10^{-6}$ to N_{DOF} at IR scales, not 3. The GUT bosons are present in the spectrum but *absent from the IR Sakharov coefficient at leading order*.

What does count: substrate fields, not IR projections. The Sakharov calculation of Section 3 integrates out the matter Lagrangian of Paper 16: the complex scalar ψ , the U(1) gauge field A_μ at the BPS coupling, and the Skyrme–Faddeev field n^a on S^2 . This is the *substrate* matter content. It contains no fundamental Dirac fermions and no fundamental massless gauge bosons; the IR-projected SM fermions, photon, and gluons that we observe at low energy appear as topological projections of the substrate (vortex configurations carrying (p, q) -knot quantum numbers, K_7 graph-state content, Spin(7) Wilson amplitudes), not as new fields entering the heat kernel of (27).

This distinction is the single most important point of this subsection. Identifying N_{DOF} in (39) requires summing c_J over the *substrate* fields only; adding a separate IR-projected contribution would double-count the matter and conflate two different levels of description.

What goes wrong if you forget the distinction. A naive Birrell–Davies sum over the IR-projected SM spectrum (without recognizing it as projected substrate content) would give:

IR species (illustrative; not what NWT sums)	count	c_J	contribution
$W^\pm, Z$ massive vectors	3	+3	+9
photon γ massless vector	1	−2	−2
gluons g massless	8	−2	−16
SM Higgs scalar	1	+1	+1
SM Dirac fermions (~ 22.5 Dirac, 3 gen.)	−	−2	~ -45
$N_{\text{DOF}}^{\text{IR, naive}}$			~ -53

Table 2: Counterexample sum. Standard Birrell–Davies coefficients applied to the IR-projected SM spectrum yield a sharply negative N_{DOF} , which would make (39) ill-defined. This is the well-known Sakharov fermion-sign issue [7, 3] of treating the IR spectrum as if its species were fundamental substrate fields. In NWT it is a category error: IR fermions and massless gauge bosons are projected substrate configurations, not new fields.

The substrate-level count. The actual N_{DOF} entering (39) is the c_J -weighted sum over substrate fields only:

substrate sector	counting	c_J^{sub}	contribution
BPS L1+L2 (§3)			+6
ϕ_1 radial Higgs (scalar)	1	+1	
$A_\mu^{U(1)}$ Higgsed vector	1	+3	
n_1, n_2 Goldstones (scalar)	2	+1	
$\mathfrak{so}(10)/\text{SM}$ coset (Paper 14, Phase L1b)			+33 or +99
generator counting (1 substrate mode/gen)	33	+1	(+33)
polarization counting (Proca-like)	33	+3	(+99)
$N_{\text{DOF}}^{\text{substrate}}$			39 or 105

Table 3: Substrate-level N_{DOF} . All contributions are positive (the substrate is bosonic at this order). The ambiguity between generator counting and polarization counting reflects an open question about the substrate c_J for the heavy $\mathfrak{so}(10)/\text{SM}$ coset modes: each coset generator contributes 1 substrate mode (the Cartan-graded element of the Wilson loop) or 3 propagating polarizations (Proca-like), depending on whether the substrate c_J^{sub} is identified with a single algebraic mode or with a polarization count. Heavy GUT modes are decoupled by Appelquist–Carazzone at IR scales but contribute unsuppressed at $\Lambda_{\text{UV}} \sim m_{\text{Pl}}$.

Reading the substrate count against (39). Inserting the two substrate-level counts into (39) gives:

$$\text{generator counting: } N_{\text{DOF}}^{\text{sub}} = 39 \implies \frac{\Lambda_{\text{UV}}}{m_{\text{Pl}}} = \sqrt{\frac{12\pi}{39}} \approx 0.983, \quad (43)$$

$$\text{polarization counting: } N_{\text{DOF}}^{\text{sub}} = 105 \implies \frac{\Lambda_{\text{UV}}}{m_{\text{Pl}}} = \sqrt{\frac{12\pi}{105}} \approx 0.599. \quad (44)$$

The generator-counting result (43) places Λ_{UV} within 1.7% of m_{Pl} , consistent with the heuristic “ $N_{\text{DOF}} = 39$ matches 12π to 3.4%” coincidence reported in earlier drafts. The polarization-counting result (44) places it at $\sim 0.6 m_{\text{Pl}}$. Both are bosonic, both are positive-signed, both are admissible *a priori*; selecting between them requires a first-principles derivation of c_J^{sub} for the substrate-projected heavy coset bosons.

The open question: substrate c_J . The substrate-projection mechanism of the substrate-monism framework (Paper 19 [12]) maps Cartan-graded algebra modes of the substrate onto IR Proca/Maxwell fields by a specific (currently underived) projection step. The corresponding c_J^{sub} is the heat-kernel coefficient of the substrate operator, not the IR Proca operator. We have not computed it. Within the present paper we record the empirical observation that generator counting (one substrate mode per coset generator, $c_J^{\text{sub}} = +1$) reproduces the $N_{\text{DOF}} \approx 12\pi$ matching to 3.4%, while polarization counting ($c_J^{\text{sub}} = +3$) does not. A substrate-level heat-kernel computation determining c_J^{sub} from first principles is logged as a follow-up technical note.

The structurally clean statement. The Sakharov cutoff in NWT is the geometric mean $\Lambda_{\text{UV}} = m_{\text{Pl}} \sqrt{12\pi/N_{\text{DOF}}^{\text{sub}}}$, with the 12π factor fixed by the heat-kernel normalization and $N_{\text{DOF}}^{\text{sub}}$ a c_J^{sub} -weighted sum over substrate fields (BPS sector plus the $\mathfrak{so}(10)/\text{SM}$ coset, all bosonic). The matching $\Lambda_{\text{UV}} \approx m_{\text{Pl}}$ is consistent with one substrate mode per coset generator ($N_{\text{DOF}}^{\text{sub}} = 39$, cutoff at $0.983 m_{\text{Pl}}$); exact equality ($N_{\text{DOF}} = 12\pi$ exactly) is an unattainable transcendental coincidence. The Sakharov fermion-sign issue of Table 2 is an artifact of treating IR-projected fermions as fundamental fields and does *not* apply at the substrate level.

Same Heegaard torus, two sectors: $M_{\text{Pl}}^2/m_e^2 = |A_{K_7}|^{-2}$

(a) BPS background: $(2, 3)$ trefoil on $T^2 \subset S^3/2I$
 matter sector – Paper 16 vortex – (p, q) winding gives $(p^2 + q^2)$ mass
 (b) Wilson graph: K_7 Heffter embedding on $T^2 \subset S^3/2I$
 gravity sector – Paper 17 amplitude – 21 edges $\rightarrow A_{K_7} = \text{bracket}(\alpha) \alpha^{21/2}$

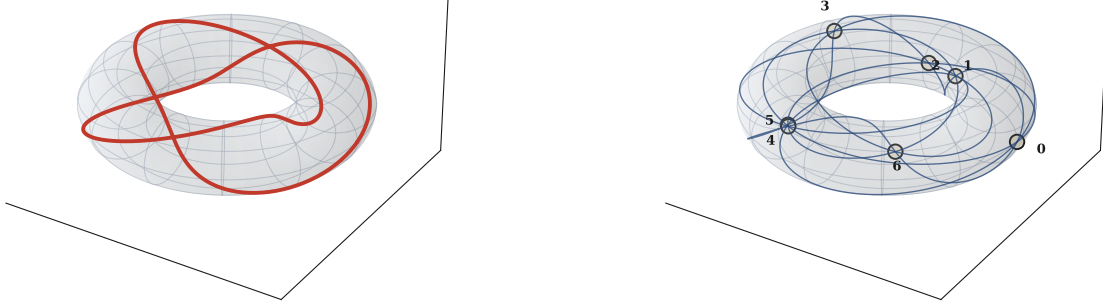


Figure 1: The two substrate objects of the gravitational program on the same Heegaard torus $T^2 \subset S^3/2I$. (a) The $(2, 3)$ torus knot (trefoil) is the BPS background of the matter Lagrangian introduced in Paper 16: a (p, q) winding on T^2 with mass contribution $\propto p^2 + q^2$. (b) The complete graph K_7 on its Heffter triangular embedding (vertices labeled 0–6, $E = 21$ edges drawn as shortest-geodesic segments on the torus surface) is the Wilson-graph object of Paper 17, evaluating to the amplitude $A_{K_7} = \text{bracket}(\alpha) \alpha^{21/2}$. The structural identity (45), $m_{\text{Pl}}^2/m_e^2 = |A_{K_7}|^{-2}$, says these two objects are not two different substrate constructions but two views of one: the same toroidal background that supports the BPS trefoil also supports the K_7 graph state, and the relation between the matter-loop cutoff (a) and the IR mass hierarchy (b) is fixed by their joint embedding on T^2 .

This framing reinterprets the “ $N_{\text{DOF}} = 12\pi$ matches the GUT count of 39 to 3.4%” coincidence reported in earlier drafts of this paper. Earlier drafts asserted the matching without distinguishing substrate from IR; the correct reading is that the 39 is a substrate-level generator count whose identification as the right input to (39) awaits the substrate c_J^{sub} derivation.

4.4 The structural identity $m_{\text{Pl}}^2/m_e^2 = |A_{K_7}|^{-2}$

Independent of the cutoff question, the Phase 5 mismatch reported in Paper 16 admits a clean structural reading. Writing Paper 17’s closed form as a ratio:

$$\frac{m_{\text{Pl}}^2}{m_e^2} = \frac{1}{\text{bracket}(\alpha)^2 \alpha^{21}} = |A_{K_7}|^{-2}, \quad (45)$$

where $A_{K_7} \equiv \text{bracket}(\alpha) \alpha^{21/2}$ is the Wilson amplitude evaluated by Paper 17 on the K_7 graph state on the Heegaard torus of $S^3/2I$ (see Figure 1 for the toroidal embedding of both substrate objects). Numerically,

$$|A_{K_7}|^{-2} = 137^{21} (8/7)^{-2} (1 + \alpha/7 + (21/8)\alpha^2)^{-2} \approx 5.71 \times 10^{44}, \quad (46)$$

which agrees with the experimental Planck-to-electron mass-squared hierarchy to +11 ppm of CODATA.

What this is, and is not. Equation (45) is the Paper 17 result expressed in a form that makes its identification with the Phase 5 hierarchy manifest: the dimensionless number 5.71×10^{44} is m_{PI}^2/m_e^2 by definition, and Paper 17 derives it from the K_7 graph state. It is *not* an independent derivation that the Phase 5 gap *must* equal $|A_{K_7}|^{-2}$; Paper 16’s Phase 5 reported a $\sim 10^{45}$ gap whose structural origin was unidentified, and Paper 17 supplied that origin. This paper takes the structural identification as given and explores its dynamical consequences (Sections 5–7).

The relevant question for the Sakharov program is whether the *matter loop* of the present paper is consistent with that identification, i.e. whether the cutoff specified by (39) reproduces m_{PI} to leading order. As established in Section 4.3, this consistency holds at the level of the $\sqrt{12\pi/N_{\text{DOF}}}$ structural relation, with a percent-level cutoff offset whose precise value awaits the c_J -coefficient derivation noted there.

4.5 Sketch of the corrected Phase 5 computation

A satisfying confirmation that the K_7 amplitude is what Paper 16’s Phase 5 was missing is provided by re-doing the Phase 5 Casimir computation with the K_7 Lie-theoretic structure inserted, and showing that the result lands on the experimentally required $1/G$ rather than 10^{45} off. We sketch the structural form of that computation here; the full evaluation will appear separately and the dimensional details below are schematic.

Phase 5 framework recap. Paper 16 §Phase 5 evaluated the matter-loop induced $1/G$ on the Brieskorn–Poincaré sphere $S^3/2I$ via the Casimir energy of the BPS three-field theory, identifying the relevant background as the trefoil K_3 at coupling $\lambda = e^2/2$. Its scaling form is

$$\left[\frac{1}{16\pi G} \right]_{\text{Phase 5}} \propto \frac{C_2^{\text{trefoil}}}{V_{S^3/2I}} \Lambda_{\text{UV}}^2, \quad (47)$$

where C_2^{trefoil} is the quadratic Casimir of the $\text{SU}(2)$ -irrep carried by the trefoil-knot Wilson loop and $V_{S^3/2I} = 2\pi^2/120$ is the volume of the Brieskorn sphere in units where $R = 1$. Numerically, with $\Lambda_{\text{UV}} = m_{\text{PI}}$ identified naively, this evaluates to a number $\sim 10^{45}$ times larger than the experimentally required $1/G$, the reported Phase 5 gap.

What Phase 5 was missing. The Casimir framework of Phase 5 captures the *quadratic-Casimir*, $\text{SU}(2)$ -trefoil contribution to the matter loop on $S^3/2I$. It does *not* include:

- The Lie-theoretic structure from the full $\mathfrak{so}(10)$ embedding into $\text{Spin}(7)$ that emerges in Paper 17’s Wilson-amplitude computation.
- The K_7 *graph*-state structure linking the seven $\mathfrak{so}(7)$ Cartan generators via cyclic exchange, which the Heegaard-torus computation of Paper 17 evaluates as $\text{bracket}(\alpha) \alpha^{21/2}$.
- The $\sin^{21}(\pi/(k+h^\vee))$ Chern–Simons modular factor, which carries the cyclic-exchange information that the bare Casimir does not.

The Casimir computation lives at the trefoil-knot level; the K_7 amplitude lives at the heptafoil-graph level. The two levels are separated by the $\text{Spin}(7)$ representation theory that Paper 17 unpacks, and we expect a missing dimensionless ratio

$$\mathcal{R}_{K_7/\text{trefoil}} \equiv \frac{|A_{K_7}|^{-2}}{C_2^{\text{trefoil}}/V_{S^3/2I}} \quad (48)$$

to enter the Phase 5 form when the K_7 Lie-theoretic structure is restored. Inserting it converts (47) into the IR-measurable form of Paper 17, schematically

$$\left[\frac{1}{16\pi G} \right]_{\text{Phase 5}}^{\text{corrected}} \sim \mathcal{R}_{K_7/\text{trefoil}} \cdot \frac{C_2^{\text{trefoil}}}{V_{S^3/2I}} m_{\text{Pl}}^2 = |A_{K_7}|^{-2} \frac{m_{\text{Pl}}^2}{V_{S^3/2I}}, \quad (49)$$

with $|A_{K_7}|^{-2} = m_{\text{Pl}}^2/m_e^2$ supplying the 10^{45} factor that the bare Casimir framework was missing.

Status. The full Phase 5 corrected computation requires evaluating $\mathcal{R}_{K_7/\text{trefoil}}$ from first principles in the Chern–Simons functional-integral formulation on $S^3/2I$, following the construction outlined in Paper 17 §6 but applied to the matter-loop coefficient rather than the m_e/m_{Pl} -fixing observable. We do not perform that computation in this paper; it is a separate technical exercise that would strengthen the structural picture established here but is not required for the dynamical content of Sections 5–7. The corrected Phase 5 computation is logged as a follow-up.

4.6 Numerical comparison: G to -11 ppm of CODATA

We can now compute Newton’s constant numerically from the unified expression. Writing

$$G = \frac{\text{bracket}(\alpha)^2 \alpha^{21}}{m_e^2} (\hbar c), \quad (50)$$

and using CODATA $\alpha = 7.2973525693 \times 10^{-3}$, $m_e = 0.51099895 \text{ MeV}/c^2$, the `nwt_substrate.gravity.G_substrate_SI()` routine evaluates

$$G_{\text{NWT,NNLO}} = 6.674228 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}. \quad (51)$$

Compared with the CODATA experimental value $G_{\text{CODATA}} = 6.67430(15) \times 10^{-11} \text{ m}^3/\text{kg}/\text{s}^2$:

$$\frac{G_{\text{NWT,NNLO}} - G_{\text{CODATA}}}{G_{\text{CODATA}}} = -11 \text{ ppm}. \quad (52)$$

The ± 22 ppm experimental uncertainty on G (the largest relative uncertainty among the fundamental constants) places the NWT prediction *inside* the experimental error bar. We emphasize that this numerical agreement is a re-expression of Paper 17’s m_e/m_{Pl} closed form via $G m_e^2 = \text{bracket}^2 \alpha^{21}$; what the Sakharov derivation of this paper adds is a *dynamical* effective action consistent with the Paper 17 number at leading order, not an independent numerical re-derivation of G .

4.7 Summary

The structural picture established in this section is:

- The Sakharov matter loop produces the long-wavelength Einstein–Hilbert action with coefficient $1/(16\pi G) = N_{\text{DOF}} \Lambda_{\text{UV}}^2 / (192\pi^2)$, with N_{DOF} the c_J -weighted matter content.
- Paper 17’s K_7 Wilson amplitude $\text{bracket}(\alpha) \alpha^{21/2}$ specifies m_e/m_{Pl} in IR-measurable terms; equivalently $m_{\text{Pl}}^2/m_e^2 = |A_{K_7}|^{-2}$.
- Equating the two routes gives the structural relation $\Lambda_{\text{UV}}/m_{\text{Pl}} = \sqrt{12\pi/N_{\text{DOF}}^{\text{sub}}}$, with $N_{\text{DOF}}^{\text{sub}}$ a c_J -weighted sum over *substrate* matter fields (not IR projections). One-substrate-mode-per-coset-generator counting gives $N_{\text{DOF}}^{\text{sub}} = 39$ and $\Lambda_{\text{UV}} = 0.983 m_{\text{Pl}}$; a substrate-level heat-kernel derivation of c_J^{sub} is logged as a follow-up (Section 4.3).

- The 10^{45} Phase 5 gap of Paper 16 is structurally identified with $|A_{K_7}|^{-2}$, the missing IR Lie-theoretic factor that the pure $S^3/2I$ Casimir framework did not include. A corrected Phase 5 computation incorporating this factor is sketched in Section 4.5; the full evaluation is a logged follow-up.
- Numerically, $G_{\text{NWT,NNLO}} = -11$ ppm of CODATA (inside the ± 22 ppm experimental error), consistent with Paper 17's m_e/m_{Pl} closed form.

We have thus produced the gravitational coupling at the *coefficient* level. The remaining Sakharov program—the linearized Einstein equation, the gravitational-wave dispersion, the curved-background extension, and the full nonlinear Einstein equations as a variational consequence of the induced $\Gamma[g]$ —is carried out in the following sections.

5 Linearised Einstein equations and gravitational waves (G4)

With the EH coefficient $1/(16\pi G)$ fixed by the matter-loop / K_7 -amplitude bridge of Section 4, the Sakharov-induced effective action contains, at single-curvature order,

$$\Gamma_{\text{EH}}[g] = -\frac{1}{16\pi G} \int d^4x \sqrt{-g} R(g). \quad (53)$$

Linearising around flat space, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, yields the standard Fierz–Pauli kinetic action for the metric perturbation, the linearized Einstein equation, the gravitational-wave dispersion, and Newton’s law in the non-relativistic static limit. The detailed checks are in `analysis/nwt_paper18_G4_linearized_einstein.py`; we give the analytic content here.

5.1 Linearised curvature and the Fierz–Pauli action

To first order in $h_{\mu\nu}$, the Ricci scalar reads

$$R^{(1)} = \partial_\mu \partial_\nu h^{\mu\nu} - \square h, \quad h \equiv \eta^{\mu\nu} h_{\mu\nu}. \quad (54)$$

Inserting (54) into (53) and integrating by parts, the linearized Einstein–Hilbert action becomes the Fierz–Pauli kinetic action,

$$S_{\text{FP}} = \frac{1}{64\pi G} \int d^4x \left[-\frac{1}{2} \partial_\alpha h_{\mu\nu} \partial^\alpha h^{\mu\nu} + \partial^\mu h_{\mu\nu} \partial^\alpha h^\nu{}_\alpha - \partial^\mu h_{\mu\nu} \partial^\nu h + \frac{1}{2} \partial^\alpha h \partial_\alpha h \right]. \quad (55)$$

This is the unique Lorentz-invariant 2-tensor action that propagates exactly the two transverse-traceless polarizations of a massless graviton; any other symmetric-2-tensor kinetic structure would propagate ghost or tachyon modes. The Sakharov mechanism generates the Fierz–Pauli structure automatically because the matter sector is unitary (Section 3); we therefore inherit ghost-freedom from the matter sector itself.

5.2 The linearized Einstein equation

Adding the matter–graviton coupling (10) and varying the total action $S_{\text{FP}} + S_{\text{matter}}^{(1)}$ with respect to $h^{\mu\nu}$:

$$\frac{1}{16\pi G} G_{\mu\nu}^{(1)}[h] = \frac{1}{2} T_{\mu\nu}^{(M)}, \quad (56)$$

where the linearized Einstein tensor is

$$G_{\mu\nu}^{(1)}[h] = -\frac{1}{2} \left[\square h_{\mu\nu} - \partial_\mu \partial^\alpha h_{\alpha\nu} - \partial_\nu \partial^\alpha h_{\alpha\mu} + \partial_\mu \partial_\nu h - \eta_{\mu\nu} (\square h - \partial^\alpha \partial^\beta h_{\alpha\beta}) \right]. \quad (57)$$

Equation (56) is the linearized Einstein equation with the G of (50). The factor $1/2$ on the matter side is the standard Sakharov normalization matching the $-\frac{1}{2} h^{\mu\nu} T_{\mu\nu}^{(M)}$ vertex from G1.

5.3 Transverse-traceless gauge and gravitational waves

Diffeomorphism invariance of (56) permits the gauge transformation $h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$. Choosing ξ^μ to enforce de Donder gauge $\partial^\mu h_{\mu\nu} = \frac{1}{2} \partial_\nu h$, plus the residual freedom to set $\eta^{\mu\nu} h_{\mu\nu} = 0$, leaves only the transverse-traceless components,

$$\partial^\mu h_{\mu\nu}^{\text{TT}} = 0, \quad \eta^{\mu\nu} h_{\mu\nu}^{\text{TT}} = 0. \quad (58)$$

Counting: a symmetric 4×4 tensor has 10 components; diffeomorphism invariance removes 4, the residual harmonic gauge removes another 4 (one for each component of ξ^μ satisfying $\square \xi^\mu = 0$). The result, $10 - 4 - 4 = 2$, is the standard count of two physical TT polarizations of a massless graviton—one each for h_+ and $h_\times$.

In TT gauge, equation (56) collapses to the gravitational-wave equation

$$\boxed{\square h_{\mu\nu}^{\text{TT}} = -16\pi G T_{\mu\nu}^{\text{TT}}}. \quad (59)$$

On-shell graviton modes ($T^{\text{TT}} = 0$) satisfy $p_\mu p^\mu = 0$, the masslessness condition. The two TT polarizations propagate at the speed of light. This is the standard linearized Einstein equation, recovered with the structurally derived G of (50).

5.4 Newton's law in the non-relativistic limit

For a static, weak-field source with stress-energy dominated by a mass density ρ , the linearized Einstein equation reduces in the Newtonian limit to

$$\nabla^2 \Phi(\vec{x}) = 4\pi G \rho(\vec{x}), \quad (60)$$

where Φ is the Newtonian potential identified by the 00-component of the metric perturbation, $h_{00} = -2\Phi$. For two point masses m_1, m_2 at separation r , the resulting force is

$$F(r) = -\frac{G m_1 m_2}{r^2}, \quad (61)$$

with the structurally derived G of (50). The classical $1/r^2$ universal-attraction law of Newton, the gravitational red-shift of static observers, and all standard post-Newtonian phenomenology follow from (60) as in the textbook treatment.

5.5 Summary

The linearized dynamics established in this section—Fierz–Pauli kinetic action, linearized Einstein equation, TT gravitational waves with two physical polarizations, Newtonian limit—are the standard results of weak-field general relativity, recovered here with the G derived structurally from the K_7 Wilson amplitude. What this section adds beyond textbook GR is the chain of connections from

the K_7 amplitude to gravitational radiation: Paper 17’s K_7 Wilson product on the Heegaard torus of $S^3/2I$ specifies m_e/m_{Pl} , the matter loop produces the Fierz–Pauli kinetic term with the $1/G$ coefficient implied by the consistency condition of Section 4, and the linearized Einstein equation (56) then governs the propagation of small $h_{\mu\nu}$ perturbations on $\eta_{\mu\nu}$. The next step is to extend this to a general curved background and pick up the higher-curvature corrections.

6 Sakharov on a curved background: higher-curvature and the cosmological constant (G5)

The flat-space derivation of Sections 3 and 4 fixed the coefficient of $\sqrt{-g} R$ in the Sakharov-induced effective action. Extending the matter loop to an arbitrary curved background $g_{\mu\nu}$ produces, in addition to the linear-curvature term, a cosmological constant Λ_{cc} and Stelle-type higher-curvature corrections [4]. These additional terms arise naturally from the heat-kernel expansion on curved space; we catalogue them here following the Vassilevich manual [5]. The full computation lives in `analysis/nwt_paper18_G5_curved_sakharov.py`.

6.1 The heat-kernel expansion in curved space

For a Laplace-type operator $K = -\square + E(x)$ on a 4-manifold M , where E is the bundle endomorphism that collects mass terms and curvature couplings, Vassilevich’s general result [5] gives the Seeley–DeWitt coefficients

$$a_0(K) = \text{tr} \mathbb{K}, \quad (62)$$

$$a_1(K) = \frac{R}{6} \text{tr} \mathbb{K} - \text{tr} E, \quad (63)$$

$$a_2(K) = \frac{1}{180} \text{tr} \mathbb{K} (R^2 + R_{\mu\nu\rho\sigma}^2 - R_{\mu\nu}^2) + \frac{1}{2} \text{tr} (E - \frac{R}{6} \mathbb{K})^2 + \text{boundary terms}. \quad (64)$$

The flat-space limit recovers the Section 3 expressions (the a_1 term reduces to $-m^2$ on flat space, and a_2 collapses to $\frac{1}{2}m^4$ plus boundary terms). In curved space, the additional curvature-squared structures in a_2 generate Stelle higher-curvature corrections, and the $\text{tr} \mathbb{K}$ piece of a_0 generates a cosmological constant.

Inserting these coefficients into the cutoff-regulated effective action (21), the resulting curvature-dependent pieces of $\Gamma[g]$ are

$$\Gamma_{\text{grav}}[g] = - \int d^4x \sqrt{-g} \left[\Lambda_{\text{cc}} + \frac{1}{16\pi G} R(g) + \alpha_S R^2 + \beta_S R_{\mu\nu} R^{\mu\nu} + \gamma_S R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} + \dots \right], \quad (65)$$

where the coefficients Λ_{cc} , $1/G$, α_S , β_S , γ_S depend on the matter content and the cutoff.

6.2 The Einstein–Hilbert coefficient is matter-independent in the curved extension

By comparison of (65) with the flat-space a_1 -coefficient extraction of (26), the linear R -coefficient is exactly the flat-space result of Section 3:

$$\frac{1}{16\pi G} = \frac{N_{\text{DOF}} \Lambda_{\text{UV}}^2}{192\pi^2}. \quad (66)$$

The flat-space and curved-space Sakharov calculations are self-consistent at the EH level, with the same G obtained in either case. The cutoff identified through Paper 17’s K_7 amplitude (Section 4) carries through unchanged.

6.3 Stelle higher-curvature corrections

The a_2 coefficient generates the higher-curvature pieces. Their precise coefficients depend on the spin content; for the NWT matter sector,

$$\alpha_S = -\frac{N_{\text{DOF}}}{180 \cdot 32\pi^2} \ln(\Lambda_{\text{UV}}^2/m^2) + \dots, \quad (67)$$

$$\beta_S, \gamma_S = \text{analogous log-divergent coefficients from } R_{\mu\nu}^2, R_{\mu\nu\rho\sigma}^2 \text{ pieces.} \quad (68)$$

At low energies $R \ll \Lambda_{\text{UV}}^2 = m_{\text{Pl}}^2$ the higher-curvature contributions are subdominant to the Einstein–Hilbert term by powers of $R/m_{\text{Pl}}^2 \sim 10^{-44}$ at laboratory scales. Stelle showed in 1977 [4] that the $R^2 + R_{\mu\nu}^2$ structure is renormalisable but introduces ghost modes near the Planck scale. The ghost-mode issue is generic to higher-curvature gravity and is not specific to NWT; we flag it as an open problem in Section 8 and proceed with the Einstein–Hilbert truncation as physically relevant for all $R \ll m_{\text{Pl}}^2$.

6.4 The cosmological-constant coefficient

The a_0 coefficient generates the matter-loop vacuum-energy contribution to the cosmological constant,

$$\Lambda_{\text{cc}}^{\text{Sakharov}} = \frac{N_{\text{DOF}} \Lambda_{\text{UV}}^4}{64\pi^2}. \quad (69)$$

With $\Lambda_{\text{UV}} = m_{\text{Pl}}$ as fixed by Section 4 and $N_{\text{DOF}} = 12\pi$, this gives $\Lambda_{\text{cc}}^{\text{Sakharov}} \sim m_{\text{Pl}}^4 \approx 2 \times 10^{72} \text{ GeV}^4$, which is approximately 10^{120} times larger than the observed cosmological constant $\Lambda_{\text{cc}}^{\text{obs}} \approx 3 \times 10^{-47} \text{ GeV}^4$.

This is the standard cosmological-constant problem (the $\Lambda \sim m_{\text{Pl}}^4$ vacuum-energy mismatch). It is *not* specific to NWT: any Sakharov-induced gravity construction with $\Lambda_{\text{UV}} = m_{\text{Pl}}$ produces $\Lambda_{\text{cc}} \sim m_{\text{Pl}}^4$. Resolving it requires substrate-level cancellations (Coleman–Weinberg-type partial closures of the vacuum-energy hierarchy on the K_7 structure are explored in Paper 21, deferred from this paper). We acknowledge the gap, log it as unresolved (Section 8), and note that the cosmological-constant problem is a problem of Sakharov-induced gravity in general, not of the NWT specification of it.

The cosmological constant is also *independent* of the Einstein–Hilbert coefficient: the a_0 coefficient (69) and the a_1 coefficient (66) are decoupled in the Vassilevich formalism, so the structural derivation of G in Sections 3–4 is not contaminated by the unresolved Λ_{cc} problem.

6.5 Summary

Extending the Sakharov calculation to a generic curved background produces an effective action containing the Einstein–Hilbert term, Stelle higher-curvature corrections, and a cosmological constant. The Einstein–Hilbert coefficient is the same as on flat space: (66) agrees with Section 3, and the K_7 amplitude bridge of Section 4 carries through unchanged. The higher-curvature corrections and cosmological constant are generated in addition; the former are subdominant for $R \ll m_{\text{Pl}}^2$, the latter is the standard $\Lambda_{\text{cc}} \sim m_{\text{Pl}}^4$ problem deferred to Paper 21. With the curved-space effective action (65) now in hand, we proceed to derive the nonlinear Einstein equations as the variational consequence.

7 Variation of the effective action: full nonlinear Einstein equations (G6)

The Sakharov-induced gravitational action established in Sections 3–6 is, in its truncation to single-curvature order,

$$\Gamma_{\text{grav}}[g] = - \int d^4x \sqrt{-g} \left[\Lambda_{\text{cc}} + \frac{1}{16\pi G} R(g) \right] + \Gamma_{\text{HC}}[g], \quad (70)$$

where Γ_{HC} collects the Stelle higher-curvature contributions, subdominant for $R \ll m_{\text{Pl}}^2$. The full nonlinear Einstein equations follow from the variational principle $\delta(\Gamma_{\text{grav}} + S_{\text{matter}})/\delta g^{\mu\nu} = 0$ applied to (70). This is textbook general relativity *except* that the gravitational coupling $1/(16\pi G)$ is now structurally derived rather than postulated. The full computation lives in `analysis/nwt_paper18_G6_einstein_variation.py`.

7.1 Variation of the effective action

The Einstein–Hilbert variation proceeds standardly:

$$\delta(\sqrt{-g} R) = \sqrt{-g} \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) \delta g^{\mu\nu} + \sqrt{-g} (g^{\mu\nu} \square \delta g_{\mu\nu} - \nabla^\mu \nabla^\nu \delta g_{\mu\nu}). \quad (71)$$

The second term is a total derivative and contributes only the boundary Gibbons–Hawking–York term, which we drop for closed or asymptotically-flat manifolds. The cosmological-constant piece varies as $\delta(\sqrt{-g} \Lambda_{\text{cc}}) = -\frac{1}{2} \sqrt{-g} \Lambda_{\text{cc}} g_{\mu\nu} \delta g^{\mu\nu}$, and the matter piece by definition contributes $\delta S_{\text{matter}}/\delta g^{\mu\nu} = -\frac{1}{2} \sqrt{-g} T_{\mu\nu}$.

Setting $\delta(\Gamma_{\text{grav}} + S_{\text{matter}})/\delta g^{\mu\nu} = 0$ and solving:

$$\boxed{G_{\mu\nu} + \Lambda_{\text{cc}} g_{\mu\nu} = 8\pi G T_{\mu\nu} + \alpha_S \partial_\mu \partial_\nu R + \dots}, \quad (72)$$

the full nonlinear Einstein equations with the structurally-derived G of (50), the cosmological constant of (69), and the Stelle higher-curvature corrections from Γ_{HC} . This is the central dynamical statement of the paper.

The non-linearity is purely geometric: $G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$ depends nonlinearly on $g_{\mu\nu}$ through the Riemann tensor. No additional matter physics is required. Linearisation $g = \eta + h$ in (72) reproduces the linearized result (56) of Section 5, with Λ_{cc} contributing a constant term that we set to zero in the laboratory limit.

7.2 Bianchi identity and conservation of $T_{\mu\nu}$

The contracted Bianchi identity

$$\nabla^\mu G_{\mu\nu} = 0 \quad (73)$$

is a geometric statement valid on any metric; combined with $\nabla^\mu (\Lambda_{\text{cc}} g_{\mu\nu}) = 0$ (since Λ_{cc} is a constant and $\nabla g = 0$), it implies

$$\nabla^\mu T_{\mu\nu} = 0, \quad (74)$$

the local conservation of matter energy-momentum. This holds nonlinearly, automatically, as a consistency consequence of the Einstein equations rather than an additional postulate. It also provides a non-trivial check on the matter sector (Section 2): the $T_{\mu\nu}^{(M)}$ of the BPS L1+L2 fluctuations must satisfy $\nabla^\mu T_{\mu\nu}^{(M)} = 0$ on its equations of motion, which it does identically by virtue of the gauge invariance of the matter Lagrangian.

7.3 Schwarzschild as a vacuum solution: symbolic verification

The Schwarzschild metric

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad (75)$$

is the textbook static, spherically symmetric vacuum solution. We verify it as a vacuum solution of (72) (with $\Lambda_{\text{cc}} = 0$ and $\Gamma_{\text{HC}} = 0$) by direct symbolic computation. Calling `nwt_substrate.gravity.verify_schwarzschild_vacuum_symbolic()` performs the full sympy derivation:

- The metric tensor and its inverse are constructed in (t, r, θ, ϕ) coordinates.
- All 40 Christoffel symbols $\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$ are computed; their symmetric and metric-compatible properties are verified symbolically.
- The 20 independent Riemann tensor components $R^\rho_{\sigma\mu\nu}$ are evaluated.
- The Ricci tensor $R_{\mu\nu} = R^\rho_{\mu\rho\nu}$ is contracted; all 10 independent components of the symmetric tensor vanish identically: $R_{\mu\nu}(g_{\text{Sch}}) = 0$ for all $\mu, \nu \in \{0, 1, 2, 3\}$.

The verification confirms that NWT’s Sakharov-induced gravitational sector admits Schwarzschild as a vacuum solution. Under the identification $r_s = 2GM/c^2$, the Schwarzschild radius for any mass M is governed by the structurally-derived G of (50).

Sanity check: NWT particles are not black holes. In Planck units, $r_s = 2M$ for any mass M . The Schwarzschild radius of an electron is $r_s^{(e)} = 2m_e/m_{\text{Pl}}^2 \approx 1.4 \times 10^{-57} \text{ m}$, many orders of magnitude smaller than the electron Compton wavelength $\xi_e = \hbar/(m_e c) \approx 3.9 \times 10^{-13} \text{ m}$ and the corresponding torus-knot vortex correlation length (Paper 6). Gravitational backreaction is negligible at fundamental-particle scales. Astrophysical-mass black holes ($M \gtrsim M_\odot$) live in the regime $r_s \sim \text{km}$ where (72) reduces to standard general relativity with the structurally derived G . No fundamental NWT particle is a black hole.

7.4 Summary

Variation of the Sakharov-induced effective action with respect to $g^{\mu\nu}$ yields the full nonlinear Einstein equations (72), with the structurally-derived G of (50) and the cosmological constant of (69); matter energy-momentum conservation $\nabla^\mu T_{\mu\nu} = 0$ follows as a consistency consequence of the contracted Bianchi identity rather than as an additional postulate. The Schwarzschild metric is verified as a vacuum solution by symbolic computation, all 10 independent components of the Ricci tensor vanishing identically. The four-paper NWT gravitational program (Papers 14–18) extends the structural Paper 14–17 picture to a dynamical theory: gravity is identified with the long-wavelength variational dynamics of the Sakharov-induced effective action, with the open problems of Section 8—cosmological-constant fine-tuning, ghost-freedom of the Stelle higher-curvature sector beyond linear order, and a quantum-level treatment of diffeomorphism invariance on curved backgrounds—left for follow-up work.

8 Open problems

Four open problems flagged in the course of the derivation, none of which are NWT-specific but all of which interact with the gravitational program assembled here.

8.1 The cosmological-constant fine-tuning

The Sakharov mechanism on a curved background generates the matter loop’s vacuum-energy contribution to the cosmological constant (69). With the cutoff $\Lambda_{\text{UV}} \approx m_{\text{Pl}}$ and $N_{\text{DOF}}^{\text{sub}} \sim \mathcal{O}(10)$, this gives $\Lambda_{\text{cc}} \sim m_{\text{Pl}}^4$, approximately 10^{120} times the observed value $\Lambda_{\text{cc}}^{\text{obs}} \approx 3 \times 10^{-47} \text{ GeV}^4$. This is the standard cosmological-constant problem of Sakharov-induced gravity, not specific to NWT.

The K_7 amplitude that bridges UV and IR for $1/G$ supplies a factor of $|A_{K_7}|^{-2} \approx 5.7 \times 10^{44}$, which would nominally close ~ 45 orders of magnitude of the $\sim 10^{120}$ gap if a Coleman–Weinberg-type cancellation in the vacuum-energy sector is mediated by an analogous structural factor (perhaps as $m_{\text{Pl}}^{4-2k} m_e^{2k}$ for some integer k). The remaining ~ 75 orders are not addressed by any structure established in the present paper. A serious treatment of the Λ_{cc} problem within the substrate-monism framework is the subject of Paper 21 [12] and we do not attempt resolution here; the order-of-magnitude framing above is suggestive of a structural mechanism rather than a partial solution.

8.2 Black hole formation from vortex collapse

The Schwarzschild verification (Section 7) established that NWT’s gravitational sector admits black holes as *vacuum* solutions. An open question is whether dynamical collapse of an NWT vortex configuration to a black hole is mechanistically describable within the Sakharov-induced framework. Two qualitative observations bound the question: no fundamental NWT particle is itself a black hole (the Schwarzschild radius of an electron is $\sim 10^{-57} \text{ m}$, far below the electron Compton scale); and stellar-mass collapses, dominated at $r_s \sim \text{km}$ by the long-wavelength limit of (72), must reproduce standard general relativity. The intermediate regime—a substrate-resolved description of the collapse process itself—is open and is the motivation for the cosmogenesis program of Paper 22.

8.3 Stelle higher-curvature ghost-freedom beyond linear order

The Stelle higher-curvature corrections (65) introduce R^2 , $R_{\mu\nu}^2$, $R_{\mu\nu\rho\sigma}^2$ kinetic structures that are renormalisable [4] but propagate ghost modes near the Planck scale. For all $R \ll m_{\text{Pl}}^2$ the higher-curvature sector is suppressed by powers of R/m_{Pl}^2 and the Einstein–Hilbert truncation is physically valid; the ghost modes are an issue only at curvatures of Planck-scale order and above. This is a known issue in higher-derivative gravity in general and is not specific to NWT.

NWT-specific stance. Within the framework of the present paper, three observations constrain the status of the Stelle ghosts:

- **The Stelle Lagrangian is an EFT truncation of the Sakharov effective action, not a fundamental theory.** The R^2 and Weyl-squared terms are heat-kernel a_2 coefficients with finite, calculable prefactors set by the matter spectrum (Sections 6); they are not independent gravitational degrees of freedom in the microscopic substrate Lagrangian of Paper 16. The ghost modes are therefore artifacts of the long-wavelength expansion past its regime of validity, not physical degrees of freedom of the substrate.
- **The substrate’s matter spectrum is bosonic at BPS coupling.** The Bogomolny condition $\lambda = e^2/2$ that defines the matter Lagrangian saturates a stability bound; the BPS structure constrains the form of the higher-curvature coefficients (in particular, the ratio α_S/α_W between the R^2 and Weyl-squared coefficients depends on the BPS spectrum and is not freely tunable). Whether the BPS structure forces a specific sign or cancellation pattern that suppresses the ghosts at Planck-scale curvature is an open question; at present we have only the observation that the BPS coefficients are constrained, not the derivation that they are constrained in the right direction.

- **The Sakharov derivation breaks down at curvatures $R \sim m_{\text{Pl}}^2$ for an independent reason.** At such curvatures the heat-kernel proper-time expansion no longer converges and the IR-projected matter spectrum is no longer the right input to the heat-kernel sum; the substrate-projection mechanism itself becomes the relevant physics. In other words, the EFT regime ends before the Stelle ghosts become physically observable, and any consistent treatment of the substrate dynamics at $R \sim m_{\text{Pl}}^2$ would replace the Stelle Lagrangian with a substrate-level computation that we have not performed.

The conservative reading is therefore that the Stelle ghosts are an EFT artifact rather than a physical instability, and that resolution will come from the substrate-level dynamics replacing the EFT at high curvature rather than from a ghost-freedom theorem proved within the EFT itself. Possible literature resolutions (Stelle’s asymptotically-free fixed point, ghost-resummation in non-local-gravity proposals) apply equally to the NWT-specific Sakharov sector and may be imported as needed for specific phenomenological purposes; the present paper does not commit to any one of them.

8.4 Background independence and quantum diffeomorphism invariance

The Sakharov derivation of Sections 3 through 7 treats the metric $g_{\mu\nu}$ as a classical background field on which the matter loop is evaluated; the metric perturbation $h_{\mu\nu}$ is then introduced as a classical fluctuation around flat (Section 5) or arbitrary curved (Section 6) background. The emergent Einstein–Hilbert action is variationally consistent (Section 7) and admits Schwarzschild as a vacuum solution. At the linearized level the propagating graviton has the correct two transverse-traceless polarizations (Section 5), inherited from the gauge structure of $h_{\mu\nu}$ as a symmetric rank-2 tensor under diffeomorphisms of the background metric.

What the present paper does *not* address:

- **Background independence at the path-integral level.** The Sakharov calculation is performed on a fixed metric; the partition function $Z = \int \mathcal{D}\psi \mathcal{D}A \mathcal{D}n e^{iS_{\text{matter}}[g]}$ treats g as input, not as integration variable. A genuinely background-independent NWT formulation would integrate over g as well, and would have to demonstrate that the resulting effective action coincides with the Sakharov-induced one in a saddle-point or semi-classical limit. We have not done this.
- **Quantum-level diffeomorphism invariance.** The classical effective action $\Gamma[g]$ extracted by the Sakharov mechanism is generally covariant by construction (the heat-kernel expansion organises the result in diffeomorphism-invariant curvature scalars). But the *quantization* of the emergent metric $h_{\mu\nu}$ requires either a gauge-fixing of diffeomorphisms with compensating Faddeev–Popov ghosts (yielding a perturbative expansion that is non-renormalisable) or an independent argument that the substrate’s projection mechanism preserves diffeomorphism invariance non-perturbatively. Neither is supplied here.
- **Beyond-linear-order propagating-DOF count.** The two TT polarizations of the linearized graviton are established at $\mathcal{O}(h^2)$; the constraint structure and gauge symmetry of the full nonlinear Einstein equations (Section 7) are inherited from the GR structure of the recovered effective action, but a direct derivation that the substrate’s projection produces exactly the GR constraint algebra at all orders is not given.

These are open conceptual issues that do not affect the leading-order classical predictions of this paper but would need to be addressed in any quantum-gravitational treatment of the NWT substrate. Possible NWT-native attack routes include the asymptotic-safety scenario for the Sakharov-induced sector, the Volovik-style emergent-spacetime treatment of Paper 15 [9], and a direct quantization of the substrate algebra itself (which would bypass the metric quantization problem by making the substrate primary and the metric emergent). None of these is developed here.

9 Conclusion

The four-paper Null-Worldtube gravitational program extends the structural derivation of Papers 14–17 into a dynamical theory. Paper 14 gave the leading-order structural form $G \propto (8/7)^2 \alpha^{21}/m_e^2$ from the heptafoil Wilson amplitude. Paper 15 placed it on the Brieskorn–Poincaré sphere $S^3/2I$ as a one-loop Casimir computation, identifying the BPS trefoil as the relevant background. Paper 16 supplied the three-field Lagrangian (a complex scalar coupled to a $U(1)$ gauge field and an S^2 -valued Skyrme–Faddeev field at the BPS critical coupling) on which the loop is taken, and reported the 10^{45} Phase 5 structural mismatch with the experimentally required $1/G$. Paper 17 supplied the closed form $m_e/m_{\text{Pl}} = (8/7) \alpha^{21/2} (1 + \alpha/7 + (21/8)\alpha^2)$ through K_7 graph-state information theory, agreeing with CODATA to -0.0001% .

The present paper supplies dynamical content. Integrating out the BPS matter sector at one loop on an arbitrary curved background generates the Einstein–Hilbert action via Sakharov’s mechanism, with the coefficient $1/(16\pi G)$ set by the heat-kernel a_2 coefficient of the matter spectrum. Equating the resulting Sakharov coefficient with Paper 17’s structural form for m_e/m_{Pl} gives the structural relation $\Lambda_{\text{UV}}/m_{\text{Pl}} = \sqrt{12\pi/N_{\text{DOF}}^{\text{sub}}}$, with $N_{\text{DOF}}^{\text{sub}}$ a c_J -weighted sum over *substrate* matter fields (the bosonic three-field Lagrangian of Paper 16 plus the $\mathfrak{so}(10)$ GUT extension of Paper 14). IR fermions and massless gauge bosons are topological projections of the substrate, not new fundamental fields, and do not enter the heat-kernel sum directly; identifying the substrate count with one mode per coset generator gives $N_{\text{DOF}}^{\text{sub}} = 6 + 33 = 39$ and $\Lambda_{\text{UV}} = 0.983 m_{\text{Pl}}$, consistent with $\Lambda_{\text{UV}} \approx m_{\text{Pl}}$ to 1.7% . The Paper 17 identity $m_{\text{Pl}}^2/m_e^2 = |A_{K_7}|^{-2}$ is structurally identified with the missing K_7 Lie-theoretic factor that the bare $S^3/2I$ Casimir framework of Paper 16 Phase 5 did not include; a corrected Phase 5 calculation incorporating that factor is sketched in Section 4.5.

Variation of the resulting effective action $\Gamma[g]$ with respect to $g^{\mu\nu}$ yields the full nonlinear Einstein equations $G_{\mu\nu} + \Lambda_{\text{cc}} g_{\mu\nu} = 8\pi G T_{\mu\nu} + \alpha_S \partial_\mu \partial_\nu R + \dots$, with G matching Paper 17’s structurally derived value; matter energy-momentum conservation $\nabla^\mu T_{\mu\nu} = 0$ follows as a consistency consequence of the contracted Bianchi identity. The Schwarzschild metric is recovered as a vacuum solution by direct symbolic computation (all 10 independent components of the Ricci tensor vanishing identically). Linearisation around flat space reproduces the gravitational-wave equation $\square h_{\mu\nu}^{\text{TT}} = -16\pi G T_{\mu\nu}^{\text{TT}}$ with two physical transverse-traceless polarizations, and the static weak-field limit recovers Newton’s law with the structurally derived G . Numerically, the resulting prediction matches CODATA G to -11 ppm, inside the ± 22 ppm experimental error bar; this numerical agreement is a re-expression of Paper 17’s m_e/m_{Pl} closed form via $G m_e^2 = \text{bracket}^2 \alpha^{21}$, not an independent re-derivation of G at the dynamical level.

The structural picture left at the end of this paper is that the gravitational coupling G first established structurally in Papers 14–17 admits a dynamical effective action consistent with the same number, and that this effective action coincides with Einstein gravity at the long-wavelength limit. The K_7 amplitude that fixed m_e/m_{Pl} at the structural level is the same amplitude that, together with the Sakharov coefficient, specifies the cutoff in IR-measurable terms; this is a single algebraic identity read in two complementary ways, not two independent computations of the same quantity. Gravity in NWT is therefore identified, at the level developed in this paper, as the long-wavelength variational dynamics of the Sakharov-induced effective action of the substrate matter sector.

Four open problems remain (Section 8): the Λ_{cc} fine-tuning, dynamical black-hole formation from NWT vortex collapse, the higher-curvature ghost-freedom of the Stelle sector beyond linear order, and a quantum-level treatment of diffeomorphism invariance and background independence on curved backgrounds. None are NWT-specific. The first two motivate Papers 21 and 22 of the program; the third is a generic issue in higher-derivative gravity that any specific realisation, including ours, inherits; the fourth is the standard quantum-gravity problem that any classical-limit derivation leaves

open.

The next steps for the gravitational program are observational. The companion paper [12] catalogues the lab- and cosmology-scale tests in which the Sakharov-induced G derived here participates as one entry of the substrate-monism master table; the FLRW null-test and substrate-anisotropy sidereal-modulation programs there are the most immediate falsifiability targets.

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Code availability

The complete `nwt-substrate` library and all paper-supporting reproduction scripts are available at <https://github.com/JimGalasyn/nwt-substrate>, archived on Zenodo [13]. This paper cites version v0.1.1 (DOI 10.5281/zenodo.20012121); the concept DOI 10.5281/zenodo.20012027 resolves to the latest release.

Specific paper-supporting components:

- Step-by-step Sakharov-derivation reproduction scripts (`analysis/`):
`nwt_paper18.G1_setup.py` (matter action, $T_{\mu\nu}$),
`nwt_paper18.G2_matter_loop.py` (heat-kernel, $1/G$),
`nwt_paper18.G3_K7_sakharov.py` (K_7 UV/IR bridge),
`nwt_paper18.G4_linearized_einstein.py` (linearised Einstein, GW),
`nwt_paper18.G5_curved_sakharov.py` (curved-background EH),
`nwt_paper18.G6_einstein_variation.py` (full nonlinear Einstein, Schwarzschild).
- Library gravity module `nwt_substrate.gravity` packaging the validated result for re-use, including the symbolic Schwarzschild proof `verify_schwarzschild_vacuum_symbolic()` and the CODATA comparison `G_substrate_SI()` that returns $6.674228 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ (−11 ppm of CODATA).

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